

Project:

Capstone_Strum_AdamN

Licensed user:

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Calculated:

2026-05-08 12:01 PM/4.2.303

SHADOW - Main Result

Calculation: Site 99 Analysis - Real Case Flicker Map

Assumptions for shadow calculations

Maximum distance for influence

Calculate only when more than 20 % of sun is covered by the blade

Please look in WTG table

Sunshine probability S (Average daily sunshine hours) [CHARLOTTETOWN]

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
3.37	4.18	4.42	5.04	6.34	7.54	7.95	7.19	5.76	3.98	2.63	2.31

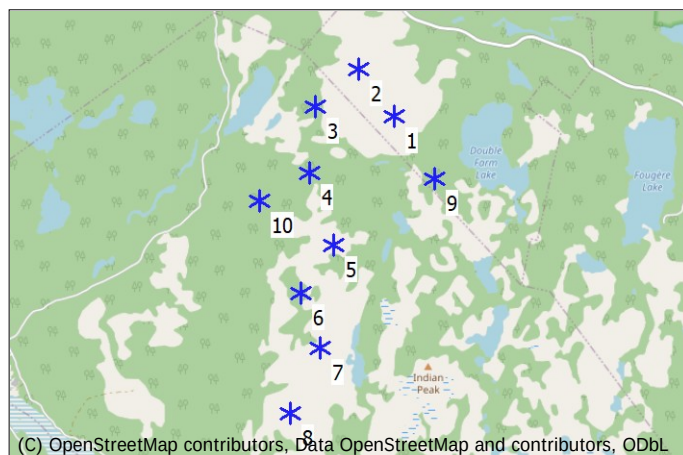
No operational time reduction. It is assumed the WTGs are always running with worst case wind direction.

A ZVI (Zones of Visual Influence) calculation is performed before flicker calculation so non visible WTG do not contribute to calculated flicker values. A WTG will be visible if it is visible from any part of the receiver window. The ZVI calculation is based on the following assumptions:

DHM: Elevation Grid Data Object: Capstone_Strum_AdamN_EMDGrid_2.wpg (14)

All coordinates are in

UTM (north)-WGS84 Zone: 20



* Existing WTG

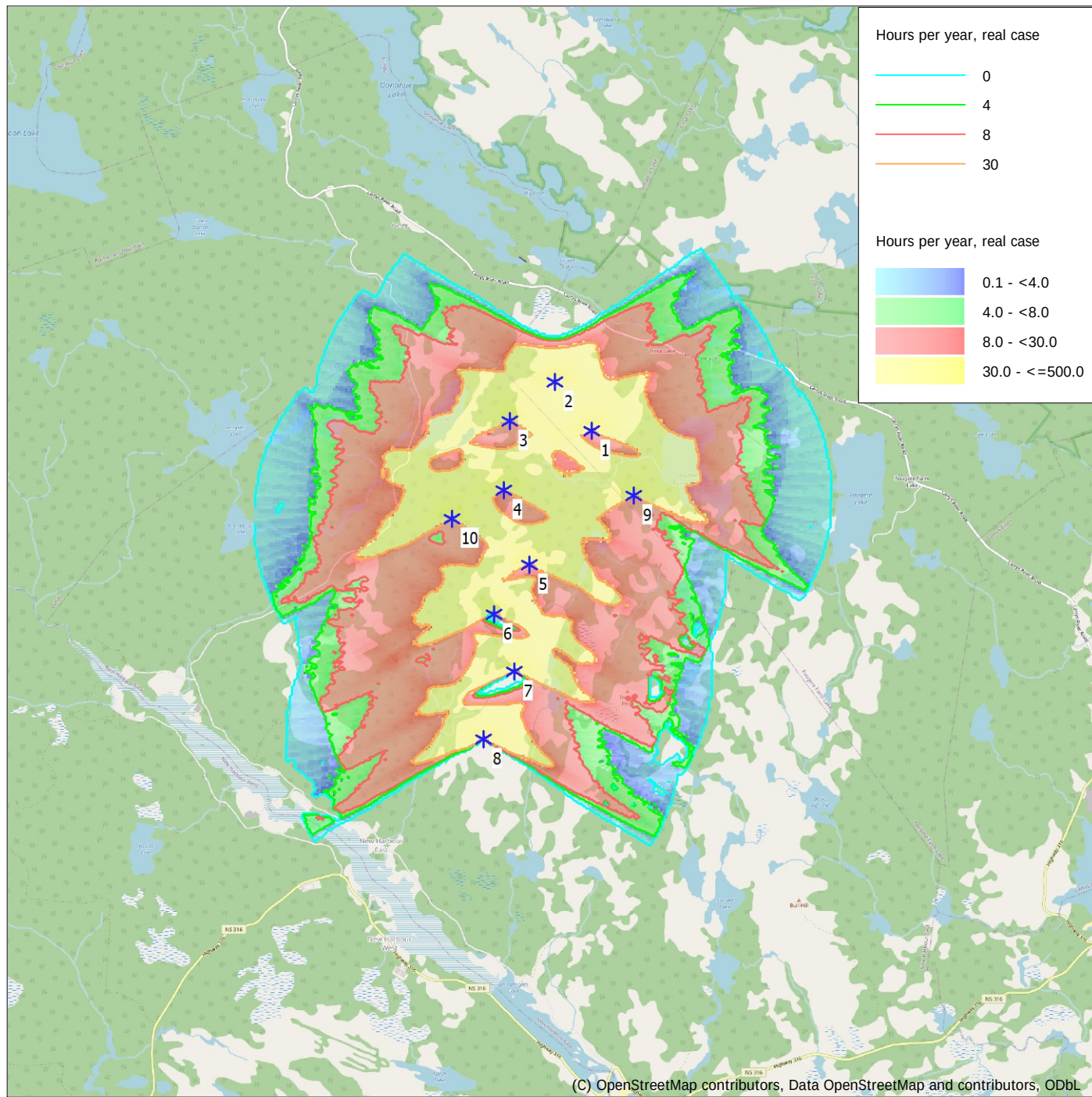
Scale 1:100,000

WTGs

	Easting	Northing	Z	Row data/Description	WTG type		Type-generator	Power, rated [kW]	Rotor diameter [m]	Hub height [m]	Shadow data	
					Valid	Manufact.					Calculation distance [m]	RPM
			[m]									[RPM]
1	620,638	5,012,469	135.0	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
2	620,158	5,013,069	120.9	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
3	619,598	5,012,569	121.0	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
4	619,538	5,011,689	132.8	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
5	619,878	5,010,749	145.0	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
6	619,438	5,010,109	140.1	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
7	619,718	5,009,389	129.9	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
8	619,338	5,008,509	120.0	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
9	621,198	5,011,649	105.9	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2
10	618,878	5,011,309	129.0	Siemens Gamesa SG 7.0-170 7000 170.0 !O!... Yes	Yes	Siemens Gamesa	SG 7.0-170-7,000	7,000	170.0	135.0	2,500	9.2

SHADOW - Map

Calculation: Site 99 Analysis - Real Case Flicker Map



0 1 2 3 4 km

Map: EMD OpenStreetMap , Print scale 1:75,000, Map center UTM (north)-WGS84 Zone: 20 East: 620,040 North: 5,010,780

* Existing WTG

Flicker map level: Elevation Grid Data Object: Capstone_Strum_AdamN_EMDGrid_2.wpg (14)
Time step: 4 minutes, Day step: 14 days, Map resolution: 30 m, Visibility resolution: 15 m, Eye height: 1.5 m